

MAT307R01 - Statistical Methods

Two Marks Questions

Measures of Dispersion, Theoretical Distributions, Large and Small Sample Tests, Non-parametric Tests, ANOVA, Correlation, Regression & SQC

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UNIT I: Measures of Dispersion and Theoretical Distributions

1. Define Range and write its formula.

Answer: Range is the difference between the maximum and minimum values in a dataset. Formula: Range = Maximum value – Minimum value. It is the simplest measure of dispersion but is affected by extreme values.

2. What is Quartile Deviation and how is it calculated?

Answer: Quartile Deviation (Q.D.) is half the difference between the third and first quartiles. Formula: $Q.D. = \frac{Q_3 - Q_1}{2}$, where Q_3 is the third quartile and Q_1 is the first quartile. It is also called semi-interquartile range.

3. Write the formula for Mean Deviation about mean and median.

Answer: Mean Deviation about mean: $M.D.(\bar{x}) = \frac{\sum |x_i - \bar{x}|}{n}$. Mean Deviation about median: $M.D.(M) = \frac{\sum |x_i - M|}{n}$, where x_i are individual observations, $\bar{x}$ is mean, M is median, and n is sample size.

4. Define Standard Deviation and write its formula for ungrouped data.

Answer: Standard Deviation (S.D.) is the square root of the arithmetic mean of squared deviations from the mean. Formula: $\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$ or $\sigma = \sqrt{\frac{\sum x_i^2}{n} - \bar{x}^2}$, where x_i are observations and $\bar{x}$ is the mean.

5. What is Coefficient of Variation and why is it useful?

Answer: Coefficient of Variation (C.V.) is the ratio of standard deviation to mean, expressed as percentage. Formula: $C.V. = \frac{\sigma}{\bar{x}} \times 100\%$. It is useful for comparing variability between datasets with different units or different means.

6. Define the r -th moment about mean and write its formula.

Answer: The r -th moment about mean is $\mu_r = \frac{\sum (x_i - \bar{x})^r}{n}$ for ungrouped data. For grouped data: $\mu_r = \frac{\sum f_i (x_i - \bar{x})^r}{N}$. The first moment about mean is always zero ($\mu_1 = 0$).

7. What is Skewness? Write any two measures of skewness.

Answer: Skewness measures the asymmetry of a distribution. Two measures are:
 (i) Pearson's coefficient: $Sk_p = \frac{\text{Mean} - \text{Mode}}{\sigma}$ or $\frac{3(\text{Mean} - \text{Median})}{\sigma}$. (ii) Bowley's coefficient: $Sk_b = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$.

8. Define Kurtosis and state the types based on kurtosis value.

Answer: Kurtosis measures the peakedness or flatness of a distribution. $\beta_2 = \frac{\mu_4}{\mu_2^2}$ is the measure. Types: (i) Mesokurtic: $\beta_2 = 3$ (normal), (ii) Leptokurtic: $\beta_2 > 3$ (peaked), (iii) Platykurtic: $\beta_2 < 3$ (flat).

9. Define a random variable and distinguish between discrete and continuous random variables.

Answer: A random variable is a function that assigns numerical values to outcomes of a random experiment. Discrete random variable takes countable values (e.g., number of heads). Continuous random variable takes any value in an interval (e.g., height, weight).

10. What is the expectation of a random variable X ? Write its formula for discrete case.

Answer: Expectation (or expected value) is the weighted average of all possible values. For discrete X : $E(X) = \sum x_i p(x_i)$, where x_i are values and $p(x_i)$ are corresponding probabilities. It represents the mean of the distribution.

11. Define Moment Generating Function (MGF) and state its importance.

Answer: The MGF of random variable X is $M_X(t) = E(e^{tX})$. For discrete: $M_X(t) = \sum e^{tx} p(x)$. For continuous: $M_X(t) = \int e^{tx} f(x) dx$. It uniquely determines the distribution and can be used to find all moments.

- 12. State the conditions for a discrete probability distribution and write the formula for Binomial distribution.**

Answer: Conditions: (i) $0 \leq p(x) \leq 1$ for all x , (ii) $\sum p(x) = 1$. Binomial distribution: $P(X = r) = {}^nC_r p^r q^{n-r}$, where n is number of trials, p is success probability, $q = 1 - p$, $r = 0, 1, 2, \dots, n$.

- 13. Write the mean and variance of Binomial distribution $B(n, p)$.**

Answer: For Binomial distribution with parameters n and p : Mean = $E(X) = np$, Variance = $\text{Var}(X) = npq$, where $q = 1 - p$. Standard deviation = $\sqrt{npq}$. The distribution is symmetric when $p = 0.5$.

- 14. State the probability mass function of Poisson distribution and its parameter.**

Answer: Poisson distribution: $P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}$, where $r = 0, 1, 2, \dots$, and $\lambda > 0$ is the parameter (average rate). Mean = λ , Variance = λ . It is used for rare events.

- 15. Write the mean and variance of Poisson distribution and state when Poisson approximates Binomial.**

Answer: For Poisson distribution with parameter λ : Mean = $E(X) = \lambda$, Variance = $\text{Var}(X) = \lambda$. Poisson approximates Binomial when n is large, p is small, and $np = \lambda$ (finite), typically $n \geq 30$ and $p \leq 0.05$.

- 16. Write the probability density function of Normal distribution $N(\mu, \sigma^2)$.**

Answer: The PDF of Normal distribution is $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, where $-\infty < x < \infty$, μ is mean, and σ^2 is variance. The curve is bell-shaped and symmetric about μ .

- 17. State the properties of Standard Normal distribution.**

Answer: Standard Normal distribution $N(0, 1)$ has: (i) Mean = 0, (ii) Variance = 1, (iii) PDF: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$, (iv) Symmetric about $z = 0$, (v) Total area under curve = 1, (vi) $P(-1.96 \leq Z \leq 1.96) = 0.95$.

- 18. How do you convert a Normal variable to Standard Normal variable?**

Answer: The transformation is $Z = \frac{X-\mu}{\sigma}$, where $X \sim N(\mu, \sigma^2)$ and $Z \sim N(0, 1)$. This is called standardization. Any probability for X can be computed using standard normal tables by converting to Z .

19. Write the probability density function of Uniform distribution $U(a, b)$.

Answer: The PDF of Uniform distribution is $f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$. Mean = $\frac{a+b}{2}$,

Variance = $\frac{(b-a)^2}{12}$. All values in $[a, b]$ are equally likely.

20. State the relationship between Binomial, Poisson, and Normal distributions.

Answer: (i) When n is large, p is small, and np is finite, Binomial $\rightarrow$ Poisson. (ii) When n is large and both np and nq are greater than 5, Binomial $\rightarrow$ Normal with $\mu = np$, $\sigma^2 = npq$. (iii) When λ is large, Poisson $\rightarrow$ Normal with $\mu = \sigma^2 = \lambda$.

UNIT II: Testing of Hypothesis

1. Write the test statistic for testing the significance of difference between sample proportion (p) and population proportion (P) for large samples.

Answer: The test statistic is $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$, where p is the sample proportion, P is the population proportion, $Q = 1 - P$, and n is the sample size.

2. What are the conditions required for applying large sample tests in hypothesis testing?

Answer: The conditions are: (i) Sample size $n \geq 30$, (ii) The sampling distribution approximately follows normal distribution, and (iii) For proportion tests, $nP \geq 5$ and $nQ \geq 5$.

3. State the formula for testing the significance of difference between two sample proportions p_1 and p_2 .

Answer: The test statistic is $z = \frac{p_1-p_2}{\sqrt{PQ\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$, where $P = \frac{n_1p_1+n_2p_2}{n_1+n_2}$ (pooled proportion) and $Q = 1 - P$.

4. How do you calculate the pooled proportion P when testing difference between two proportions?

Answer: The pooled proportion is calculated as $P = \frac{n_1p_1+n_2p_2}{n_1+n_2}$, where n_1, n_2 are sample sizes and p_1, p_2 are sample proportions respectively.

5. Write the test statistic for testing significance of difference between sample mean $\bar{x}$ and population mean μ when population variance is known.

Answer: The test statistic is $z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$, where $\bar{x}$ is sample mean, μ is population mean, σ is population standard deviation, and n is sample size.

6. State the test statistic when population variance is unknown in large sample test for single mean.

Answer: When population variance is unknown, $z = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$, where s is the sample standard deviation. For large samples, this follows approximately normal distribution.

7. Write the general test statistic for testing difference between two sample means $\bar{X}_1$ and $\bar{X}_2$.

Answer: The test statistic is $z = \frac{\bar{X}_1-\bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$, where σ_1^2, σ_2^2 are population variances and n_1, n_2 are sample sizes.

8. What is the test statistic when samples are drawn from the same population with equal variances $\sigma_1 = \sigma_2 = \sigma$?

Answer: The test statistic becomes $z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$. When σ is unknown, it is approximated by $\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$.

9. What distributions are used for small sample tests and when is each applicable?

Answer: Small sample tests use: (i) t -distribution for testing means when population is normal and variance unknown, (ii) F -distribution for comparing variances, and (iii) χ^2 -distribution for goodness of fit tests.

10. State the degrees of freedom for t -distribution in one-sample test and write its variance formula.

Answer: Degrees of freedom $\nu = n - 1$. The variance of t -distribution is $\frac{\nu}{\nu - 2}$ where $\nu > 2$, and mean = 0.

11. Write the test statistic for one-sample t -test and state its degrees of freedom.

Answer: The test statistic is $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$ with degrees of freedom $\nu = n - 1$, where s is sample standard deviation.

12. What are the key assumptions for conducting a one-sample t -test?

Answer: Assumptions are: (i) Sample is drawn from normally distributed population, (ii) Population variance is unknown, (iii) Sample size $n < 30$, and (iv) Observations are independent.

13. Write the test statistic for two-sample t -test when sample sizes are unequal ($n_1 \neq n_2$).

Answer: $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ with $\nu = n_1 + n_2 - 2$ degrees of freedom.

14. State the test statistic when sample sizes are equal ($n_1 = n_2 = n$) and samples are independent.

Answer: The test statistic is $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}}$ with degrees of freedom $\nu = 2n - 2$.

15. Write the test statistic for paired sample t -test and explain the notation.

Answer: $t = \frac{\bar{d}}{\frac{s}{\sqrt{n-1}}}$ with $\nu = n - 1$ degrees of freedom, where $\bar{d} = \frac{\sum d_i}{n}$, $d_i = x_i - y_i$, and $s^2 = \frac{\sum d_i^2}{n} - (\bar{d})^2$.

16. When do we use paired t -test instead of two-sample t -test?

Answer: Paired t -test is used when $n_1 = n_2 = n$ and samples are dependent (related), such as before-after measurements on same subjects, to control for individual variation by analyzing differences within pairs.

17. Write the test statistic for F -test to compare two population variances and state its degrees of freedom.

Answer: $F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2}$ (where $\hat{\sigma}_1^2 > \hat{\sigma}_2^2$) with degrees of freedom $(\nu_1, \nu_2) = (n_1 - 1, n_2 - 1)$, where $\hat{\sigma}^2 = \frac{n}{n-1}s^2$.

18. State the null and alternative hypotheses for F -test of equality of variances.

Answer: Null hypothesis: $H_0 : \hat{\sigma}_1^2 = \hat{\sigma}_2^2$ (no significant difference between population variances). Alternative hypothesis: $H_1 : \hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$.

19. State the critical values for different types of tests at 5% level of significance.

Answer: For large samples at $\alpha = 0.05$: Two-tailed $Z_\alpha = 1.96$, Right-tailed $Z_\alpha = 1.64$, Left-tailed $Z_\alpha = -1.64$. For small samples, use $t_\nu(\alpha)$ from t -tables.

20. Define the decision rule for hypothesis testing and explain Type I and Type II errors.

Answer: Decision rule: If $|z| \leq |z_\alpha|$, accept H_0 ; otherwise reject H_0 . Type I error: Rejecting true H_0 (probability = α). Type II error: Accepting false H_0 (probability = β).

UNIT III: Non-parametric Tests and Design of Experiments

1. What are non-parametric tests and when are they used?

Answer: Non-parametric tests are distribution-free tests that do not assume population follows a specific distribution. They are used when: (i) population distribution is unknown or non-normal, (ii) data is ordinal or ranked, (iii) sample size is small, (iv) outliers are present.

2. Write the test statistic for Chi-square goodness of fit test.

Answer: The test statistic is $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$ with degrees of freedom $\nu = k - 1 - m$, where O_i are observed frequencies, E_i are expected frequencies, k is number of classes, and m is number of parameters estimated.

3. State the null hypothesis for testing independence of attributes using Chi-square test.

Answer: Null hypothesis: H_0 : The two attributes are independent (no association between them). The test statistic is $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ with $\nu = (r - 1)(c - 1)$ degrees of freedom, where r is rows and c is columns.

4. How do you calculate expected frequencies in Chi-square test for contingency table?

Answer: Expected frequency for cell (i, j) is calculated as $E_{ij} = \frac{(\text{Row total}_i) \times (\text{Column total}_j)}{\text{Grand total}}$. The sum of expected frequencies equals the sum of observed frequencies.

5. What is Mann-Whitney U-test and when is it used?

Answer: Mann-Whitney U-test is a non-parametric test for comparing two independent samples to test whether they come from the same population. It is used as an alternative to two-sample t -test when normality assumption is violated. It tests medians of two populations.

6. Write the formula for calculating U statistic in Mann-Whitney U-test.

Answer: The U statistics are: $U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$ and $U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$, where n_1, n_2 are sample sizes, R_1, R_2 are sum of ranks. The test statistic is $U = \min(U_1, U_2)$.

7. What is Wilcoxon signed-rank test and state its purpose?

Answer: Wilcoxon signed-rank test is a non-parametric test for paired samples (or single sample against a hypothesized median). It is used as an alternative to paired t -test when normality is not satisfied. It considers both direction and magnitude of differences.

8. Explain the procedure for computing Wilcoxon signed-rank test statistic.

Answer: Steps: (i) Calculate differences $d_i = x_i - y_i$, (ii) Ignore zero differences, (iii) Rank absolute differences $|d_i|$, (iv) Assign signs to ranks, (v) Calculate T^+ = sum of positive ranks and T^- = sum of negative ranks, (vi) Test statistic $T = \min(T^+, T^-)$.

9. What is Run test and what does it test?

Answer: Run test (Wald-Wolfowitz test) is used to test randomness of a sequence. A run is a sequence of identical symbols preceded and followed by different symbols. It tests H_0 : sequence is random. If number of runs is too few or too many, reject H_0 .

10. Write the formula for expected number of runs and its variance.

Answer: For n_1 symbols of one type and n_2 of another: Expected runs = $\mu_R = \frac{2n_1n_2}{n_1+n_2} + 1$. Variance = $\sigma_R^2 = \frac{2n_1n_2(2n_1n_2-n_1-n_2)}{(n_1+n_2)^2(n_1+n_2-1)}$. For large samples, $Z = \frac{R-\mu_R}{\sigma_R}$.

11. What is Kruskal-Wallis H-test and when is it applied?

Answer: Kruskal-Wallis H-test is a non-parametric test for comparing three or more independent samples. It is used as an alternative to one-way ANOVA when normality assumption fails. It tests whether samples come from populations with equal medians.

12. Write the test statistic for Kruskal-Wallis H-test.

Answer: The test statistic is $H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$, where k is number of groups, n_i is size of i -th group, R_i is sum of ranks in i -th group, $N = \sum n_i$. H follows χ^2 with $\nu = k - 1$.

13. Define Spearman's rank correlation coefficient and write its formula.

Answer: Spearman's rank correlation coefficient (r_s) measures monotonic relationship between two variables using ranks. Formula: $r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$, where d_i is difference between ranks of paired observations and n is number of pairs. Range: $-1 \leq r_s \leq 1$.

14. What is Kendall's tau and how does it differ from Spearman's correlation?

Answer: Kendall's tau (τ) is another rank correlation coefficient based on concordant and discordant pairs. Formula: $\tau = \frac{n_c - n_d}{\frac{1}{2}n(n-1)}$, where n_c is number of concordant pairs and n_d is discordant pairs. It is more robust than Spearman's for small samples.

15. What is Design of Experiments and state its objectives?

Answer: Design of Experiments (DOE) is systematic approach to plan, conduct, and analyze experiments. Objectives: (i) Determine effect of factors on response, (ii) Control experimental error, (iii) Maximize information with minimum resources, (iv) Ensure validity and reliability of conclusions.

16. Define one-way classification and write its mathematical model.

Answer: One-way classification (One-way ANOVA) classifies observations based on one factor. Model: $x_{ij} = \mu + \alpha_i + \epsilon_{ij}$, where μ is overall mean, α_i is effect of i -th treatment, ϵ_{ij} is random error, $i = 1, 2, \dots, k$ and $j = 1, 2, \dots, n_i$.

17. Write the ANOVA table structure for one-way classification.

Answer: ANOVA Table:

Source	SS	df	MSS
Between treatments	SST	$k - 1$	$MST = SST/(k - 1)$
Within treatments	SSE	$N - k$	$MSE = SSE/(N - k)$
Total	TSS	$N - 1$	

Test statistic: $F = \frac{MST}{MSE}$.

18. Define two-way classification and state its advantages.

Answer: Two-way classification analyzes effect of two factors on response variable. Model: $x_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$. Advantages: (i) Tests two factors simultaneously, (ii) Reduces experimental error, (iii) More efficient than conducting two separate one-way experiments, (iv) Controls variation due to blocking.

19. What is Completely Randomized Design (CRD) and state its characteristics?

Answer: CRD is simplest experimental design where treatments are assigned completely at random to experimental units. Characteristics: (i) All units are homogeneous, (ii) No blocking, (iii) Any number of treatments and replications, (iv) Analyzed by one-way ANOVA. Maximum flexibility but less efficient.

20. Explain Randomized Block Design (RBD) and state when it is used.

Answer: RBD groups experimental units into homogeneous blocks; each block contains all treatments assigned randomly. Used when: (i) Experimental units are heterogeneous, (ii) Need to control one source of variation (blocking factor), (iii) Number of treatments $\leq$ units per block. More efficient than CRD.

21. What is Latin Square Design (LSD) and write its characteristics?

Answer: LSD controls two sources of variation (rows and columns); each treatment appears once in each row and column. Characteristics: (i) Number of rows = columns = treatments = n , (ii) Controls two nuisance factors, (iii) n^2 experimental units required, (iv) Very efficient but restrictive. Used when two blocking factors present.

UNIT IV: Correlation, Regression, and Statistical Quality Control

1. Define correlation and distinguish between positive and negative correlation.

Answer: Correlation measures the degree and direction of linear relationship between two variables. Positive correlation: both variables increase together ($r > 0$). Negative correlation: one variable increases while other decreases ($r < 0$). Range: $-1 \leq r \leq 1$.

2. Write Karl Pearson's formula for coefficient of correlation.

Answer: Pearson's correlation coefficient: $r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
or $r = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{[n \sum x_i^2 - (\sum x_i)^2][n \sum y_i^2 - (\sum y_i)^2]}}$.

3. State the properties of coefficient of correlation.

Answer: Properties: (i) $-1 \leq r \leq 1$, (ii) $r = 0$ implies no linear correlation, (iii) $r = \pm 1$ implies perfect linear correlation, (iv) r is unit-free and symmetric: $r_{xy} = r_{yx}$, (v) Change of origin and scale does not affect r .

4. What is rank correlation and write Spearman's formula for it?

Answer: Rank correlation measures monotonic relationship using ranks instead of actual values. Spearman's rank correlation: $r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$, where d_i is difference in ranks and n is number of pairs. Used for ordinal data or when relationship is non-linear.

5. Define regression and distinguish it from correlation.

Answer: Regression is a statistical method to predict one variable from another. Correlation measures strength and direction of relationship; regression provides equation for prediction. Correlation is symmetric ($r_{xy} = r_{yx}$); regression is not (regression of Y on X differs from X on Y).

6. Write the equations of regression lines of Y on X and X on Y .

Answer: Regression line of Y on X : $Y - \bar{Y} = b_{yx}(X - \bar{X})$ or $Y = a + b_{yx}X$, where $b_{yx} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$. Regression line of X on Y : $X - \bar{X} = b_{xy}(Y - \bar{Y})$, where $b_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(y_i - \bar{y})^2}$.

7. State the relationship between correlation coefficient and regression coefficients.

Answer: The relationship is: $r = \pm \sqrt{b_{xy} \cdot b_{yx}}$, where sign of r is same as sign of b_{xy} and b_{yx} . Also, $r = b_{yx} \cdot \frac{\sigma_x}{\sigma_y} = b_{xy} \cdot \frac{\sigma_y}{\sigma_x}$. If $b_{xy} \cdot b_{yx} > 1$, calculations are incorrect.

8. Write the properties of regression coefficients.

Answer: Properties: (i) b_{xy} and b_{yx} have same sign, (ii) $|b_{xy} \cdot b_{yx}| \leq 1$, (iii) If $r = 0$, both $b_{xy} = 0$ and $b_{yx} = 0$, (iv) Regression coefficients are not symmetric: $b_{xy} \neq b_{yx}$, (v) Both regression lines pass through $(\bar{x}, \bar{y})$.

9. What is the angle between two regression lines?

Answer: The angle θ between regression lines is given by $\tan \theta = \frac{(1-r^2)\sigma_x\sigma_y}{r(\sigma_x^2 + \sigma_y^2)}$, where $\sigma_x^2 + \sigma_y^2 \neq 0$. When $r = \pm 1$, lines coincide ($\theta = 0$). When $r = 0$, lines are perpendicular ($\theta = 90$).

10. Define Statistical Quality Control (SQC) and state its objectives.

Answer: SQC is application of statistical methods to control and improve quality of products/processes. Objectives: (i) Detect and eliminate assignable causes of variation, (ii) Maintain process in control, (iii) Reduce defects and variability, (iv) Achieve continuous improvement, (v) Ensure product meets specifications.

11. Distinguish between control charts for variables and control charts for attributes.

Answer: Control charts for variables are used for measurable quality characteristics (length, weight, temperature) using $\bar{X}$ -chart and R -chart. Control charts for attributes are used for countable characteristics (defective/non-defective, number of defects) using p -chart, np -chart, c -chart.

12. Write the control limits for $\bar{X}$ -chart (mean chart).

Answer: For $\bar{X}$ -chart: Central Line (CL) = $\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}$, Upper Control Limit (UCL) = $\bar{\bar{X}} + A_2 \bar{R}$, Lower Control Limit (LCL) = $\bar{\bar{X}} - A_2 \bar{R}$, where k is number of samples, $\bar{R}$ is average range, and A_2 is constant from table.

13. Write the control limits for R -chart (range chart).

Answer: For R -chart: Central Line (CL) = $\bar{R} = \frac{\sum R_i}{k}$, Upper Control Limit (UCL) = $D_4 \bar{R}$, Lower Control Limit (LCL) = $D_3 \bar{R}$, where R_i is range of i -th sample, k is number of samples, and D_3, D_4 are constants from table.

14. State the purpose of using $\bar{X}$ -chart and R -chart together.

Answer: $\bar{X}$ -chart monitors process average (central tendency) to detect shifts in process mean. R -chart monitors process variability (dispersion) to detect changes in process variation. Both are used together because a process is in control only when both mean and variability are stable.

15. Write the control limits for p -chart (proportion defective chart).

Answer: For p -chart: Central Line (CL) = $\bar{p} = \frac{\sum p_i}{k} = \frac{\text{Total defectives}}{\text{Total inspected}}$, UCL = $\bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$, LCL = $\bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$, where n is sample size and k is number of samples.

16. What is np -chart and when is it preferred over p -chart?

Answer: np -chart monitors number of defectives (not proportion) in samples of constant size n . Control limits: CL = $n\bar{p}$, UCL = $n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$, LCL = $n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$. Preferred when sample size is constant as it's easier to plot actual counts.

17. Write the control limits for c -chart (number of defects chart).

Answer: For c -chart: Central Line (CL) = $\bar{c} = \frac{\sum c_i}{k}$, UCL = $\bar{c} + 3\sqrt{\bar{c}}$, LCL = $\bar{c} - 3\sqrt{\bar{c}}$, where c_i is number of defects in i -th unit and k is number of units inspected. Used when counting defects per unit (area of opportunity).

18. Distinguish between defective and defect in quality control terminology.

Answer: Defective: An item that fails to meet specifications and is rejected as a whole (unit is either defective or not). Used in p -chart and np -chart. Defect: A single non-conformity in an item; an item may have multiple defects but still be acceptable. Used in c -chart.

19. What are the two types of causes of variation in a process?

Answer: (i) Chance/Random/Common causes: Natural inherent variations that cannot be eliminated (always present), produce stable pattern. (ii) Assignable/Special causes: Specific identifiable variations that can be detected and eliminated (not always present), cause process to go out of control. SQC aims to eliminate assignable causes.

20. Define the term "process in statistical control".

Answer: A process is in statistical control when it operates with only chance causes of variation present, showing stable and predictable pattern. Characteristics: (i) All sample points fall within control limits, (ii) No trends or patterns in plotted points, (iii) Points randomly distributed around center line, (iv) Process is stable and predictable.

Key Formulas Summary

Unit I - Measures of Dispersion and Distributions

- Standard Deviation: $\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$
- Coefficient of Variation: $C.V. = \frac{\sigma}{\bar{x}} \times 100\%$
- Binomial: $P(X = r) = {}^nC_r p^r q^{n-r}$, Mean = np , Variance = npq
- Poisson: $P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}$, Mean = Variance = λ
- Normal: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Unit II - Testing of Hypothesis

- Large sample proportion: $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$
- Difference of proportions: $z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$
- Large sample mean: $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$
- Small sample mean: $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$
- F -test: $F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2}$ where $\hat{\sigma}^2 = \frac{n}{n-1} s^2$
- Paired t -test: $t = \frac{\bar{d}}{\frac{s}{\sqrt{n-1}}}$

Unit III - Non-parametric Tests

- Chi-square: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$
- Mann-Whitney: $U = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$
- Spearman: $r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$
- Kruskal-Wallis: $H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$

Unit IV - Correlation, Regression & SQC

- Correlation: $r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
- Regression: $r = \pm \sqrt{b_{xy} \cdot b_{yx}}$
- $\bar{X}$ -chart: $UCL = \bar{\bar{X}} + A_2 \bar{R}$, $LCL = \bar{\bar{X}} - A_2 \bar{R}$
- R -chart: $UCL = D_4 \bar{R}$, $LCL = D_3 \bar{R}$
- p -chart: $UCL/LCL = \bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$
- c -chart: $UCL/LCL = \bar{c} \pm 3\sqrt{\bar{c}}$